Planning Hallucination Attribution

Query: Which academic publisher published the paper "On Two Mathematical Representations for Semantic Maps"?
Answer: De Gruyter

Agent Trajectory

Step 1

Here are the facts: ...
Plan:
1. Search for the paper...
prioritizing sources Elsevier...
5. select Elsevier as the academic publisher

Step 2

Web Search Tool

On Two Mathematical Representations for Semantic Maps
Elsevier
Parameters

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Step 4

The academic publisher of the paper "On Two Mathematical Representations for Semantic Maps" is Elsevier.

(Qwen3-32B) Answer:

Hallucination Judgment: **YES**
Responsible Step: **4**
Causual Explanation: The agent concluded that Elsevier is the academic publisher of the paper without sufficient evidence. The web search results did not confirm Elsevier as the publisher, and the search for the 'Journal of Mathematical Semantics' failed to provide any publisher information. The final answer was based on assumptions rather than verified data.

Ground Truth:

Hallucination Judgment: **YES**
Responsible Step: **1**
Causual Explanation: Step 1 Correctly understood the question, but guessed 'Elsevier' in advance in the task planning and wrote it into the plan, resulting in biased retrieval in the future, which belongs to Task Decomposition error and ultimately leads to incorrect answers.

Analysis:

The model correctly identifies the hallucination but misattributes its cause, blaming the final answer step rather than the earlier planning step where the error is introduced. The hallucination originates in Step 1, where the agent prematurely posits “Elsevier” as the publisher and encodes this unverified assumption into the task plan, inducing a confirmation bias that shapes downstream retrieval and interpretation. In contrast, the model focuses on Step 4, where the incorrect conclusion is stated, without tracing the causal chain back to the initial planning commitment that preconditions the trajectory.